

LUNENBURG, NS, Canada

WMO: 710415

Lat: 44.360N

Lon: 64.300W

Elev: 4

StdP: 101.28

Time Zone: -4.00 (NAA)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-11.8	-17.6	0.8	-11.0	-15.8	0.9	-9.7	16.3	1.7	14.8	0.2	6.1	310	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	24.6	18.9	23.1	18.3	21.8	18.0	20.7	23.1	19.8	21.7	19.1	20.7	5.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	14.7	21.6	19.1	13.9	20.7	18.4	13.3	20.0	59.9	23.3	56.9	21.8	54.3	20.7	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 14.1	12.3	10.9	DB	-16.5	28.2	2.8	1.5	-18.5	29.3	-20.2	30.2	-21.7	31.0	-23.8	32.1	(4)
(5)			WB	-15.7	22.2	1.9	1.3	-17.1	23.1	-18.2	23.8	-19.2	24.5	-20.6	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-2.6	-2.6	0.2	5.1	9.6	14.2	18.1	18.5	16.0	11.2	6.0	1.2	(6)
(7)		DBStd	8.22	4.92	4.38	3.90	2.90	2.69	2.71	2.07	1.99	2.76	3.41	3.77	4.36	(7)
(8)		HDD10.0	1669	391	352	305	150	41	2	0	0	0	27	128	274	(8)
(9)		HDD18.3	3848	649	585	563	398	273	126	29	22	78	222	370	532	(9)
(10)		CDD10.0	921	0	0	1	2	27	128	251	262	181	63	7	0	(10)
(11)		CDD18.3	61	0	0	0	0	1	3	22	26	9	1	0	0	(11)
(12)		CDH23.3	126	0	0	0	0	3	12	53	37	20	1	0	0	(12)
(13)		CDH26.7	16	0	0	0	0	0	1	9	2	4	0	0	0	(13)
(14)	Wind	WSAvg	5.0	5.8	5.8	5.8	5.2	4.7	4.4	3.7	3.7	4.3	5.2	5.5	5.9	(14)
(15)	Precipitation	PrecAvg	1437	137	114	125	111	110	102	94	90	113	134	151	156	(15)
(16)		PrecMax	1786	243	191	280	177	319	249	189	201	336	287	259	249	(16)
(17)		PrecMin	1072	55	47	40	45	45	29	20	45	49	39	78	70	(17)
(18)		PrecStd	164	44	40	52	34	57	49	39	40	59	57	53	49	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	8.9	13.4	15.5	22.4	25.2	28.0	26.1	25.6	21.1	15.4	12.2	(19)
(20)			MCWB	8.5	8.1	8.6	9.6	15.8	18.9	20.3	20.1	20.3	16.3	14.2	11.6	(20)
(21)		2%	DB	7.3	6.5	8.6	12.7	17.8	22.0	24.4	24.2	22.9	18.6	14.1	10.3	(21)
(22)			MCWB	6.4	5.4	5.8	8.2	12.4	15.8	18.6	19.1	19.0	16.2	12.9	9.6	(22)
(23)		5%	DB	5.7	4.7	6.6	10.8	15.3	19.9	22.8	22.9	21.2	17.3	12.8	9.1	(23)
(24)			MCWB	4.8	3.6	4.6	8.1	11.2	15.3	18.2	18.5	17.9	15.6	11.9	8.4	(24)
(25)		10%	DB	4.1	3.3	5.1	9.3	13.3	18.2	21.3	21.7	19.8	16.2	11.6	7.6	(25)
(26)			MCWB	3.0	2.2	3.3	6.9	10.3	14.7	18.0	18.4	17.4	14.6	10.6	6.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	8.6	10.4	12.1	16.8	19.7	21.6	21.7	21.6	18.5	15.0	11.9	(27)
(28)			MCD B	9.2	8.8	13.4	14.4	21.2	23.0	24.7	23.6	24.4	19.4	15.3	12.2	(28)
(29)		2%	WB	6.7	6.0	6.9	10.2	14.1	17.9	20.4	20.8	20.0	17.1	13.3	9.9	(29)
(30)			MCD B	7.2	6.4	8.0	11.1	16.5	20.7	23.3	23.0	21.7	18.0	13.8	10.2	(30)
(31)		5%	WB	5.1	4.1	5.3	8.8	12.3	16.6	19.5	20.0	19.1	16.2	12.1	8.6	(31)
(32)			MCD B	5.7	4.7	6.5	10.1	14.1	18.3	21.6	21.6	20.2	17.2	12.6	9.1	(32)
(33)		10%	WB	3.3	2.6	3.8	7.5	11.0	15.5	18.7	19.1	18.2	15.0	10.8	6.9	(33)
(34)			MCD B	4.2	3.4	5.1	8.9	12.8	17.2	20.4	20.7	19.3	16.0	11.6	7.6	(34)
(35)	Mean Daily Temperature Range	MDBR	6.4	6.3	6.0	5.7	5.8	5.9	5.5	5.9	6.0	5.8	5.6	6.0	(35)	
(36)		5% DB	MCD BR	7.8	7.3	7.7	8.5	9.5	9.3	7.9	7.5	7.3	6.8	6.6	7.4	(36)
(37)			MCWBR	8.2	7.3	5.8	5.6	5.5	5.0	4.0	3.9	4.7	5.2	6.6	8.1	(37)
(38)			5% WB	MCD BR	7.6	7.5	7.4	7.0	8.0	7.7	6.3	6.2	5.5	5.9	6.4	7.3
(39)		MCWBR		8.1	7.9	6.7	5.9	5.6	5.0	4.2	3.8	4.3	5.4	6.9	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.300	0.331	0.357	0.376	0.404	0.422	0.399	0.361	0.337	0.319	0.303	(40)	
(41)		taud	2.482	2.519	2.455	2.427	2.418	2.369	2.333	2.397	2.472	2.538	2.541	2.518	(41)	
(42)		Ebn at Noon	824	900	908	907	895	867	845	854	864	843	801	789	(42)	
(43)		Edh at Noon	70	80	97	109	113	120	123	111	95	78	66	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.44	2.31	3.51	4.33	4.98	5.47	5.45	5.05	4.10	2.65	1.52	1.13	(44)	
(45)		RadStd	0.11	0.18	0.27	0.39	0.63	0.52	0.39	0.34	0.26	0.29	0.16	0.09	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	+0.43	+0.57	N/A	N/A	+58	+8			(47)

Nomenclature: See separate page